

An Industrial Control System Physical Testbed for Naval Defense Cybersecurity Research

(PhD) Franck SICARD
Naval Group
Naval Cyber Laboratory (NCL)
Ollioules, France
franck.sicard@naval-group.com

(Eng.) Estelle HOTELLIER
Naval Group
Naval Cyber Laboratory (NCL)
Ollioules, France
estelle.hotellier@naval-group.com

(PhD) Julien FRANCO
Naval Group
Naval Cyber Laboratory (NCL)
Ollioules, France
julien.franco@naval-group.com

Abstract—Industrial Control Systems are used nowadays in many safety-critical applications, such as Naval Defense systems. These latter need high-level protection against cyberattacks, which can lead to potential disastrous consequences (e.g., components sabotage, Denial of Service, human deaths). Thus, these infrastructures, before being deployed in mission, need intensive security testing and validation. Moreover, cybersecurity research is required to anticipate future attacks. The research of new countermeasures has to be led on realistic platforms for getting precise and fruitful feedback for cyberdefense. This paper describes an Industrial Control System testbed for Naval Defense cybersecurity research. This realistic testbed implements a representative model of a warship, on which practical attacks and related countermeasures can be safely benchmarked. After describing the features of our physical testbed, we illustrate its relevance by describing four different attack scenarios. This testbed will be very useful in future works to elaborate and validate innovative cyberdefense measures (like knowledge-based and behavior-based intrusion detection) against network and physical process attacks, especially by generating representative datasets.

Index Terms—Industrial Control Systems, Testbed, Naval Defense Systems, Cybersecurity, Security Assessment.

1. Introduction

Industrial Control Systems (ICSs) ensure productivity, reliability and safety in many fields of activity such as Energy production and distribution, Manufacturing, Transportation, Chemical and Oil or even Water treatment and distribution. ICSs are critical and highly inter-connected infrastructures composed of a digital system interacting with a physical system; they are also called Cyber-Physical Systems (CPSs). NIST 800-82-r2 [1] defines these systems as a combination of control components acting together to achieve an industrial objective.

With the ever increasing inter-connectivity of industrial components, ICSs are more and more vulnerable and prone to cyberattacks. This security issue can be explained by an IT (Information Technology)/OT (Operational Technology) convergence. The main difference between IT and OT systems is found in the nature of the data they rely on. In OT, physical data is manipulated whereas in IT, data has no material meaning. Historically, ICSs were isolated

OT networks and cybersecurity was not a threat worth to consider. They are no longer isolated and they rely on IT as well, bringing new vulnerability surfaces inherited from the IT world.

Since the beginning of the 21st century, ICSs are the target of attacks taking advantage of these large and unprotected attack surfaces. It is worth citing some famous attacks that explicitly targeted ICSs, such as Stuxnet (2009 – 2011, Iran) [2], BlackEnergy (2015, Ukraine) [3] or Oldsmar Attack (2021, USA) [4]. These attacks had disastrous consequences going from sabotaging Uranium enrichment program of a country to depriving 230,000 people of electricity or attempting to poison a city water supply. Moreover, some attacks in the maritime domain started to occur as well in the last decade. Among the major attacks stand NotPetya (2017, against Maersk Group) and Ragnar Locker (2018 – 2020, against CMA-CGM company). So far, these attacks are ransomware targeting maritime ports, but more subtle attacks on ships embedded ICSs are to fear in the future. If some cyber risk management guidelines already exist for ports [5], it is not yet the case for ship cybersecurity or Naval Defense. And we can guess that the consequences of a cyberattack on such platforms would be catastrophic. For instance, Royal Navy's warship breakdown (2016, Persian Gulf) led to a complete system power failure (Denial of Service (DoS) of the propulsion and weapons)¹. Even if the root cause of the induced DoS was an intrinsic warship design problem, it is reasonable to think that a cyberattack targeting the concerned equipment could induce similar consequences.

All these attack examples emphasize the need for securing industrial systems. They show that an attack can have huge impact on physical equipment, economical and environmental aspects and even human life. This is why developing solutions directed towards ICS cybersecurity is fundamental. A security assessment platform, like a testbed, is one efficient solution to develop attack detection approaches. It is a way to reproduce a model of real industrial systems, excluding the potential dangerous physical consequences of cyberattacks. However, the state-of-the-art describes only a few initiatives dedicated to maritime and Naval Defense representative platforms, and only a few of them are focused on ICS cybersecurity. Moreover, the realism of most of the published platforms

1. <https://www.theguardian.com/uk-news/2016/jun/07/destroyers-will-break-down-if-sent-to-middle-east-admits-royal-navy>

can be discussed since they generally embed virtual ICS components, and only high-level descriptions of the platforms and considered attacks are provided.

Contributions: This paper describes an ICS physical testbed for Naval Defense cybersecurity research. Since it embeds physical ICS components, it is a high-fidelity testbed that provides demonstrative effects of cyberattacks on a warship, by possibly impacting its direction, stability, and manoeuvrability, disabling its artillery, etc. The architecture of our testbed and four cyberattack scenarios are described in details. The ultimate goal of our testbed is to extensively test and validate in safe conditions the efficiency of cyberdefense measures.

Outline: In Section 2 we provide the background for our work and give basic definitions that will further allow an easier comparison with the state-of-the-art. In Section 3 our industrial testbed is presented in details and Section 4 provides four examples of representative attacks. In Section 5 we compare our testbed with other relevant ones. We then conclude and give perspectives.

2. Background and Position of the Paper

2.1. High-Level Architecture of Ships

Ships, and especially warships, are very complex systems. (War)ships integrate multiple interdependent systems, defined as multilayered networks, which make them very hard to handle, especially for non-experts in naval defense. Only a few attempts have been made to model them, e.g. [6] recently tried to predict anomaly propagation in a naval water distribution CPS, which is only a limited (even vital) part of a ship. The authors also provided a high-level definition of on-board systems interdependencies in warships. The article defines five main (war)ship's functional block systems: *platform systems* in charge of the critical and basic ship functions (e.g., propulsion, direction, etc.), *bridge systems* in charge of surface situation awareness and navigation monitoring, *IT systems* including entertainment and 3rd-party systems (collecting useful data for maintenance, safety and security), *core IT/OT systems* specific for the (war)ship missions and *general communication systems* e.g., SATCOM, VHF, 3/4/5G, etc.

The five mentioned functional block systems interact together to form a complete industrial infrastructure. On our side, we rely on a more hierarchical view, aggregating some of the aforementioned functional blocks, which is related to the Purdue model described hereafter.

2.2. Purdue Model for ICSs

In this paper, we rely on the typical architecture for industrial infrastructures, defined in IEC 62264 standard [7]. More specifically, we refer to the Purdue model which is a hierarchical reference model showing distributed control functions. It is structured in 6 levels: Corporate Strategy (strategy, direction), Operational Management (planning), Process Management (scheduling, inter-unit management), Supervisory Control (unit management), Local Control and Process / Operative Part.

Fig. 1 illustrates the Purdue model Architecture. Thus, each level is composed of characteristic elements that

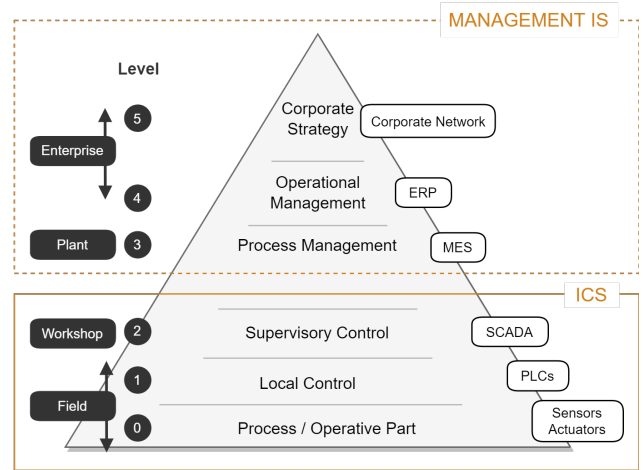


Figure 1. Illustration of Purdue Model Architecture

are indicated on the right side of the figure. On the higher levels (Levels 5 to 3), the Corporate Strategy relies on a Corporate Network, the Operational Management uses Enterprise Resource Planning (ERP) tools and the Process Management can be executed with Manufacturing Execution System (MES) or other process scheduling tools. These three levels constitute the Management Information System (MIS) of the organization. Concurrently, the lower levels of the architecture form the ICS. The Supervisory Control (Level 2) uses Supervisory Control And Data Acquisition (SCADA) equipment. The Local Control (Level 1) is made through Programmable Logic Controllers (PLCs). And the Process and Operative Part (Level 0) is composed of field devices such as sensors and actuators.

Each level exchanges information with its upper and lower levels. The higher the level, the greater the volume of data to be processed. As a consequence, the reaction time increases along with the understanding of the context. For instance, at the Local Control level, the average reaction time is in the order of a second, while that of Supervisory Control is in the order of a minute. Conversely, the Local Control level will have an understanding of a limited number of inputs/outputs, while the Supervisory Control level will drive several Local Control level devices. Furthermore, for Levels 5 to 3 of the Purdue model, we will generally speak of “deferred” time, while for Levels 2 to 0 we will speak of “real” time. The latter can be broken down into two sub-categories: “hard” for Levels 1 to 0 and “soft” for Level 2. However, in this paper, we will focus on the three lower levels (2 to 0 of Purdue Model) that mainly constitute our ICS testbed.

2.3. Testbed: Definition and Criteria

With the development of the cybersecurity issue on industrial systems, more and more ICS testbeds of these systems have been developed for testing intensively the efficiency of cyberdefense measures. Related to this topic, [8] deserves particular attention since it is a recent and complete survey on ICS testbeds for cybersecurity research. In the following, we will use intensively this reference to position our paper.

Reference [8] defines *testbed* as a testing infrastructure, consisting of a scaled-down version of a real ICS, created *ad hoc* to reproduce real-world systems but in a controlled environment. Reference [8] classifies three possible testbeds: *physical*, *virtual* and *hybrid* ICS platforms. ICS physical testbeds use real hardware and software to configure both the network and physical layers. These testbeds are the most accurate since they are the nearest to the reality. Real experimental conditions (noise, latencies, etc.) lead to a precise and detailed understanding of the system, which will allow more efficient countermeasures (e.g., low False Positive Rate). Moreover, this kind of platform allows to exploit the vulnerabilities of a specific device or protocol implementation weakness. On the contrary, virtual ICS testbeds use software simulations, and/or virtual machines, and/or emulations to replicate a system (networks and components). Even they are cheaper solutions than ICS physical testbeds, this option generates data with lower fidelity. Since Naval Defense systems are very critical and needs a high degree of modelling precision, we have chosen to implement a ICS physical testbed. Optionally, individual virtual components can be also integrated in our ICS testbed for quick prototyping (e.g., IDS, PLC), but this low degree of virtualisation does not allow to consider our ICS testbed as an hybrid one.

Testbeds must then be differentiated from two other close concepts: Cyber Ranges (CRs) and Digital Twins (DTs). CRs are mainly virtual testbeds dedicated to train operational teams to improve their responsive capacity in case of a cyber crisis, and show safely the consequences of an attack to non-experts (e.g., C-level managers). CRs allow to create topologies, scenarios, and content for feeding pre-built defensive and offensive exercises. They also embed some specific functionalities like traffic generators or Learning Management System (LMS). Clearly, the objectives of testbeds and CRs are different; in testbeds, cybersecurity experts want to experiment countermeasures on a representative of their system. On the other side, cybersecurity analysts, operators and possibly non-experts use CRs to quickly instantiate exercises of cyber attacks on a given system (possibly not the one the trainees daily operate), to increase or assess their cyber awareness level, or their operational experience. With regard to these definitions, our platform is clearly a testbed. Moreover, for sure, our hybrid testbed is not a DT. A DT can be defined as a digital representation of a physical object, asset or system (here, a ship). The traditional roles of DTs are predictive maintenance, failure analysis and recovery, and alternative architecture design testing. Since Cyber DTs are mainly virtual platforms, also coming with additional challenges [9], and are costly to create, maintain and run, our choice is directed to physical testbeds.

According to [8], testbeds can be benchmarked and compared through four main criteria: i) Fidelity, ii) Repeatability, iii) Measurement Accuracy, and iv) Safe execution. Our testbed, like the others, is a trade-off between all these criteria with respect to our constraints.

i) Fidelity. A testbed with a high degree of fidelity perfectly replicates the devices and processes from a real-world ICS. The more layers of physical, IT, and OT networks are implemented in a testbed, the more realistic it is. Introducing Cyber DTs helps us to clarify that our physical testbed is not completely identical to a specific

system of a warship, but sufficiently representative to get significant and interesting feedback. Once again, building a perfect replica of a warship from an IT/OT point of view only for cybersecurity research is unaffordable. Moreover, publishing defense-sensitive information is not permitted. Instead, we assume our choice of providing generic ship functions with widespread IT/OT components, allowing us to mount representative cyber-attacks and test our countermeasures. Examples of representative attacks which can be mounted on our physical testbed will be described in Section 4. A counterpart of the realism of our physical testbed is that it is not possible to accelerate the time (contrary to virtual platforms), so it is assumed that a long time will be needed to experiment some specific attacks.

ii) Repeatability. Repeatability of the experiments generally means higher confidence. It allows third-parties reproducing attacks and countermeasures in the same conditions than other experiments, in order to compare the findings and other solutions on the same system. By nature, it will be challenging for other researchers to perfectly reproduce our ICS physical testbed, even if we will provide all the needed details in this paper.

iii) Measurement Accuracy. In order to get accurate and reliable data from our ICS physical testbed, synchronized sensors can be integrated in different places (see Section 4). These passive probes do not disrupt the underlying physical process.

iv) Safe execution. Reference [8] also explains that it is difficult (if not impossible) to deploy attacks in a real environment since they can damage the physical process or some devices. We completely agree with that argument, and this is why, instead on working on a replica of a warship, we decided to build our ICS physical testbed, which can be viewed as a “sandboxed” environment with safe executions of attacks.

3. Presentation of our ICS Physical Testbed

When visitors enter our industrial testbed, they are immediately immersed in the context of industrial cybersecurity due to the presence of Commercial-Off-The-Shelf (COTS), real Human-Machine Interfaces (HMIs), and a small warship replica (see Fig. 2 and Fig. 3).

Our testbed implements a representative model of a surface warship. It is composed of a physical Operative part (Level 0 of the Purdue model) serving the major functions of a warship in the form of a physical model, such as Propulsion or Artillery, and a digital part grouping the functions of Local Control (Level 1 of the Purdue model) and Supervisory Control (Level 2 of the Purdue model). This environment is representative of a warship in the brands of equipment (Siemens and Schneider Electric) that compose it and in the general architecture but it is not similar in terms of complexity and quantity of equipment. The simplified architecture scheme of our testbed is available in Fig. 4. Some equipment are not represented (such as industrial firewalls connected to switches).

The architecture implemented in our testbed shows four main functions of the ship: Direction, Energy, Artillery and Propulsion. We call *domain* loop the system comprising the local network of a specific function. For each domain, Fig. 2 shows the control devices view and



Figure 2. Our Testbed (Industrial Control Devices view)

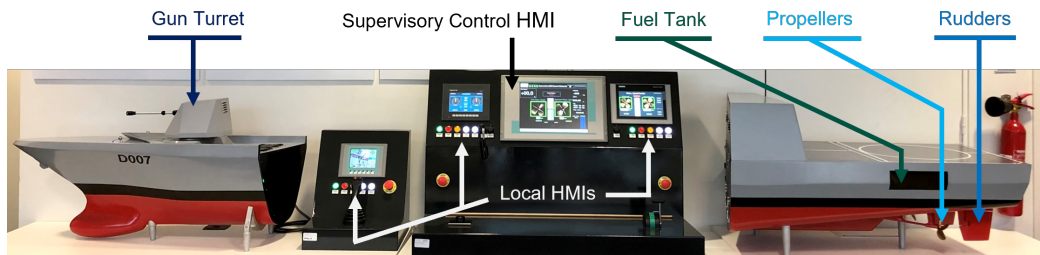


Figure 3. Our Testbed (HMIs and Physical view)

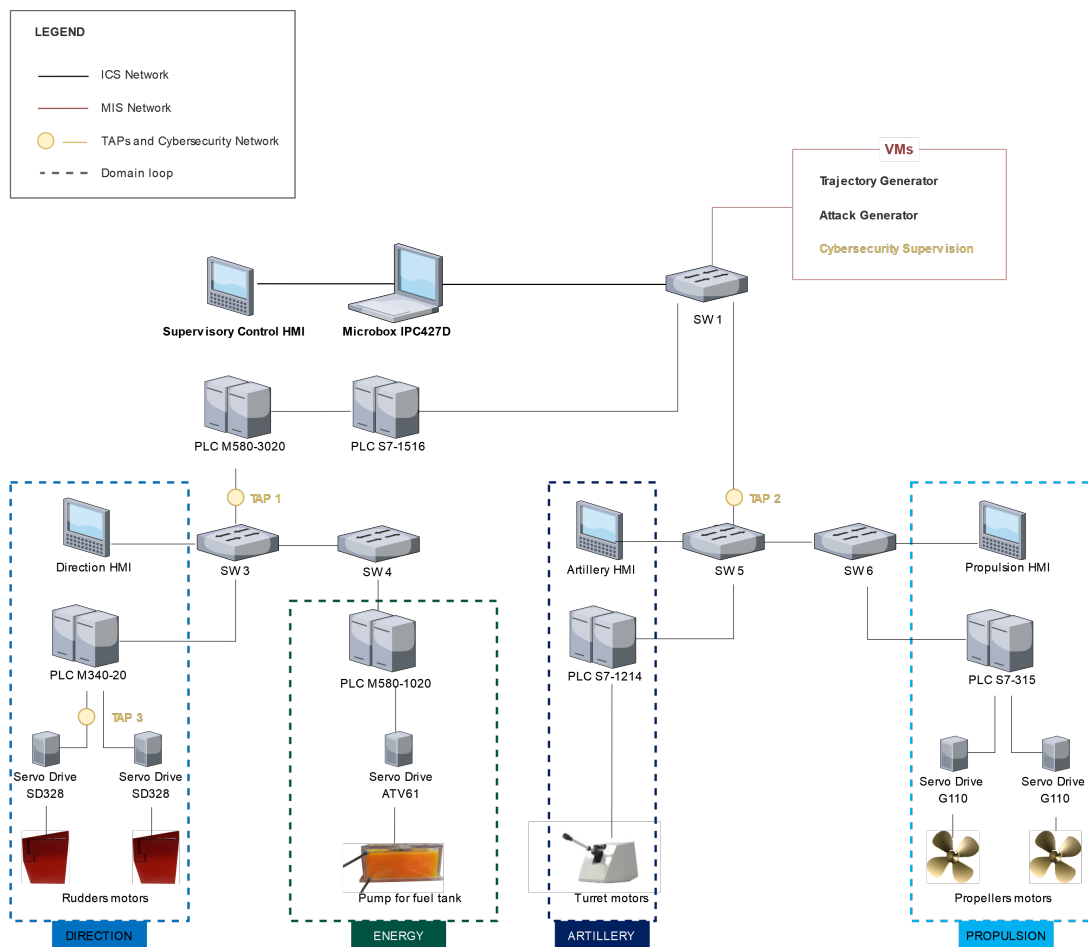


Figure 4. Simplified View of our Testbed Architecture

Fig. 3 the physical view. Four domain loops can be identified:

- Direction domain (Schneider Electric PLC M340-20): allowing to control the direction of the ship. This involves acting on the rudders position in order to vary the ship direction.
- Energy domain (Schneider Electric PLC M580-1020): allowing to supply fuel. It is a functional servitude that operates filling and emptying the fuel tank.
- Artillery domain (Siemens PLC S7-1214): allowing to take action on the 76mm gun turret model. It acts on the positioning of the gun turret (azimuth, elevation) and ammunition firing.
- Propulsion domain (Siemens PLC S7-315): allowing to control the propulsion of the ship. The aim is to act on the propulsion propellers to vary the speed of progress of the ship.

Each domain loop is driven by a control PLC. The different control PLCs communicate with each other via coordination PLCs (M580-3020 and S7-1516). These coordination PLCs are also used as relays with the Supervisory Control of our testbed. Each domain loop is controlled by authorized human operators, either manually on the equipment, locally from a local HMI or remotely from the Supervisory Control HMI.

It is important to note that these OT systems (domain loops and Supervisory Control) are complemented by an IT level corresponding to Levels 3 to 5 of the Purdue model (denoted MIS Network in Fig. 4). This IT level, detailed in Section 3.2, is composed of Virtual Machines (VMs) either generating traffic on our ICS physical testbed (“normal” traffic with the Trajectory Generator VM and “malicious” traffic with the Attack Generator VM) or managing cybersecurity information (Cybersecurity Supervision VM).

3.1. Detailed Description of our ICS Physical Testbed

3.1.1. Monitoring – SCADA. A dedicated Supervisory Control function is set up: all the monitoring and the implementation of various operations on the domains defined above are centralized (corresponding to Level 2 of Purdue model). The Supervisory Control is made possible through:

- The two aforementioned coordination PLCs (Schneider Electric M580-3020 and Siemens S7-1516) ensuring the consolidation of orders from the domain loops.
- The Siemens Microbox IPC427D (industrial computer), that implements all the Supervisory Control HMI variables. The variables are shared on the one side with the Siemens PLCs of Artillery and Propulsion domains and on the other side with the coordination PLCs serving as a Modbus/TCP to Profinet gateway towards Direction and Energy domains.
- The Supervisory Control HMI is the display for the Siemens Microbox. It represents the remote control station and allows a global monitoring of the testbed.

3.1.2. Direction Domain Loop. A dedicated Direction domain loop is set up: all the operations of starting up and operating the rudders of the ship’s model are done via this loop. The Direction domain loop is composed of:

- A control panel housing the industrial control devices (Schneider Electric PLC M340-20 and also network equipment such as switches and firewalls; corresponding to Layer 1 of Purdue model).
- A local HMI representing the Direction local control station.
- The aft part of a surface frigate containing two rudders each controlled by a stepper motor driven by two servo drives Schneider Electric SD328 (Operative Part, corresponding to Layer 0 of Purdue model).

The control of the Operative Part via the servo drives is done with a fieldbus network using Modbus RTU communication protocol (the notion of fieldbus is further detailed in Section 3.3).

3.1.3. Energy Domain Loop. A dedicated Energy domain loop is set up: it contributes to supply the Propulsion loop, and it also ensures the tank filling and emptying function. At each increase or decrease of the propulsion speed, the Energy domain regulates the fuel consumption in a proportional way. The Energy domain loop consists of:

- A control panel housing the industrial control devices (Schneider Electric PLC M580-1020 and also network equipment such as switches and firewalls; corresponding to Layer 1 of Purdue model).
- A local HMI representing the Energy local control station.
- A fuel tank located on the upper deck of the surface frigate model containing solenoid valves controlled by a peristaltic pump, itself controlled by a servo drive Altivar ATV61 (Operative Part, corresponding to Layer 0 of Purdue model).

The control of the Operative Part via the servo drive is done with a fieldbus network using Modbus RTU communication protocol.

3.1.4. Artillery Domain Loop. A dedicated Artillery domain loop is set up: it covers all the operations of starting up and controlling the 76mm gun turret model. The Artillery domain loop consists of:

- A control panel housing industrial control devices (Siemens PLC S7-1214C and also network equipment such as switches and firewalls; corresponding to Layer 1 of Purdue model).
- A local HMI representing the Artillery local control station.
- The bow of a surface frigate containing a miniature 76mm gun turret (Operative Part, corresponding to Layer 0 of Purdue model).

For this domain, the control of the gun is done via All or Nothing (AON) inputs/outputs and fieldbus (Profibus) which directly address an electronic board whose design will not be developed here.

3.1.5. Propulsion Domain Loop. A dedicated Propulsion domain loop is set up: it carries the operations of starting up and operating the propellers of a ship's model propellers. The Propulsion domain loop consists of:

- A control panel housing industrial control devices (Siemens PLC S7-315 and also network equipment such as switches and firewalls; corresponding to Layer 1 of Purdue model).
- A local HMI representing the Propulsion local control station.
- The rear part of a surface frigate containing two steel propellers, each controlled by a stepper motor driven by two servo drives Siemens G110 (Operative Part, corresponding to Layer 0 of Purdue model).

The control of the operating part via the servo drives is done with a fieldbus network using Profibus communication protocol.

3.2. Operating Data Generation

Our testbed can process data in manual, local or remote modes. However, for scalability and repeatability reasons, automatic data generation tools have been developed. Normal navigation data can be generated with the Trajectory Generator VM and data corresponding to attack scenarios can be generated with the Attack Generator VM.

The Trajectory Generator VM is a set of scripts allowing to simulate the behavior of a crew on board. Thus, normal "network" and "process" traffic is generated, creating normal navigation data. To do this, the equipment of the different local loops must be placed in remote control mode. In order to generate a trajectory, i.e. a succession of control events staggered over time, the corresponding normal navigation data file must respect the format of a key indicating the navigation sequence with the following values:

- rudder angle ("helm" key of integer type),
- duration of the order, i.e. the time that the speed order and the helm angle will be maintained ("duration" key of string type),
- speed in knots ("speed" key of character string type).

This file is in JSON format and allows to create a navigation scenario composed of one or more sub-scenarios (for example, Exit from Port, Manoeuvre 1, Manoeuvre 2, Return to Port). A script is executed in order to:

- parse the different fields of the JSON file,
- connect to the different concerned PLCs (Snap7 library for the Propulsion PLC and PyModbus library for the Direction PLC),
- check the respect of some initial parameters of the testbed (power-on bit, neutral position of the throttle, activation of the HMI in the remote mode, activation bit of the rudder and propulsion motors),
- create and start the "navigation" task (i.e. processing the normal navigation data file). It remains active until the end of the scenario or until an interruption is formulated.

This VM is hosted in a datacenter near the industrial equipment. Its interconnection has been made in a way that the orders flow as if they were coming from the Supervisory Control.

The Attack Generator VM is based on the scripts and integration philosophy of the Trajectory Generator VM. Its scripts allows the following functions:

- connect / disconnect to the concerned equipment (PLC or fieldbus, details are provided further),
- obtain information on the state of the PLC's CPU if needed,
- launch an attack by specifying the attack number, related parameters values and diverse options,
- stop the current attack.

For now, the attacks can be played on Propulsion and Direction domains. They can operate at different levels of the Purdue model. And they can impact only the Operative part or the Supervisory Control in addition of the Operative part. We present the different attacks by grouping them by scenarios:

- Scenario 1 consists in generating a new node on the testbed network, that communicates with other devices of the network,
- Scenario 2 consists in violating a security invariant of the system (for example forcing the rudders in opposition instead of putting them in the same position),
- Scenario 3 consists in performing an offset attack by adding a constant to the current command value (i.e. changing the value of a process variable),
- Scenario 4 consists in performing a "cavitation" attack. This attack allows to generate an oscillation of a configurable amplitude, on the current command value.

The functioning of the Attack Generator VM and details of the four attack scenarios are provided in Section 4.2. Note that a scenario can take diverse variations depending on the targeted domain, the network level in the architecture and other options.

3.3. Cybersecurity Components Deployed

For cybersecurity research, we also deployed a Cybersecurity Supervision VM. It is used for cybersecurity monitoring and the purpose of this VM is to detect and analyse cyber incidents. The VM aggregates and acts on logs coming from various cyber sensors. To gather information about the data flowing in our testbed, we use various Network Test Access Points (TAPs), at different levels of our architecture (represented by yellow dots in Fig. 4). The cyber sensors situated at these points, do not deteriorate network traffic and they allow a monitoring of the system at runtime. In our testbed, we are able to capture network traffic between Levels 2 and 1, and between Levels 1 and 0 of the Purdue model. Depending on where the capture is performed, different technologies and equipment are deployed. They are presented in the following paragraphs and do not appear in Fig. 4 since only the TAPs positions are indicated (numbered from 1 to 3).

Concerning the lower levels (Levels 1-0) of the Purdue model, the communication between process components and Local Control elements is made through fieldbuses. Capturing traffic on a fieldbus can be a real asset for anomaly detection since it gives access to data closest to the physical process. The notion of fieldbus characterizes a serial bidirectional communication network connecting industrial components on a unique communication medium. Fieldbus components are called nodes and they are of diverse types, such as sensors, actuators, PLCs, local controllers, remote inputs / outputs and so on. The architecture of the communication is simplified compared to traditional Ethernet-based networks. Indeed, considering the Open Systems Interconnection model (OSI model), only Layers 1 and 2 (sometimes Layer 7) are considered in fieldbuses architectures. Consequently, common tools used for Ethernet-based capture do not operate on fieldbuses.

To address this issue, we use RS-485 to USB probes together with a Modbus RTU decoder and the open-source Network IDS Zeek². The fieldbus traffic capture is implemented in the Direction domain loop of the testbed, between the PLC M340-20 and one Servo Drive SD328 (location marked TAP 3 on Fig. 4). Fig. 5 shows the elements involved in the capture and their positioning. The fieldbus protocol on this specific case is Modbus RTU. On the physical layer (Layer 1 of the OSI model), electrical signals are exchanged between nodes, following RS-485 standard detailed in [10]. Therefore the two RS-485 to USB dongles permit to decode electrical signals and pass them on to Zeek. The first dongle is used to sniff every information that flows on the bus while the other is used for sending attacks on the bus. As most of the IDS on the market, Zeek is by default restricted to Ethernet-based protocols. But it is also one of the more flexible and configurable IDS. Adding a protocol parser is a well-documented process that allows to extend Zeek framework to fieldbus protocols. Note that our implementation is developed for Modbus RTU but it would be easy to add probes for other fieldbus protocols, respecting the same framework.

On higher levels (Levels 2-1) of the Purdue model, the testbed is equipped with COTS network sensors also known as OT IDS sensors. These sensors capture traffic at the locations marked TAP 1 and 2 in Fig. 4. Our testbed integrates several OT IDS sensors solutions. We are able to compare their detection performances on the same attack scenario. The security of the testbed also includes industrial firewalls (not represented in Fig. 4) to filter communications up to the application level (Level 7 of the OSI model). Our maritime testbed is equipped with manageable switches to segment the industrial system architecture and filter communications (e.g. by MAC address).

The deployed cyber sensors (OT IDS sensors at Levels 2-1 of Purdue model and fieldbus probes Levels 1-0 of Purdue model) permit to collect numerous data of various nature. That allows to develop and test different intrusion detection methodologies such as knowledge-based and behavior-based approaches (i.e. approaches based on process data) adapted to the Naval Defense domain.

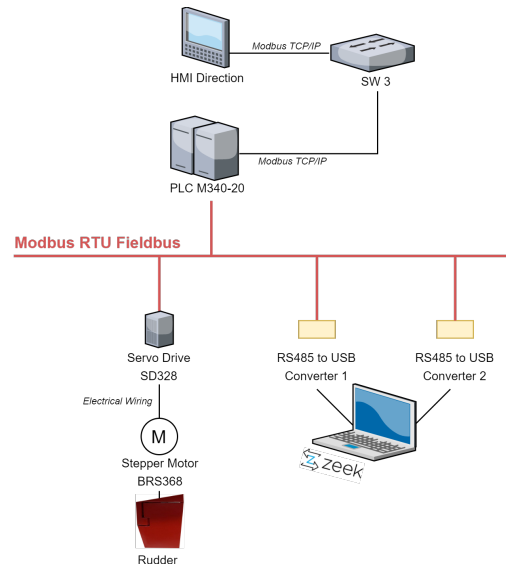


Figure 5. Modbus RTU Fieldbus Traffic Capture Framework

4. Four Examples of Representative Attacks

4.1. Normal (i.e. Benign) Scenario

In this section, we will talk exclusively about driving the warship in a normal scenario, this is done by using the Trajectory Generator VM only (we exclude driving in manual mode or via the local HMI). Thus, during this scenario, the different command events contained in the JSON file are reverberated into the different domains. At the end of the scenario, the equipment are put back to idle state for the next trajectory (rudder in zero position and propulsion equal to 0).

4.2. Scenario and Detection

Several types of attack scenarios can be executed on the testbed. However, in this paper we divided them into 2 main categories:

- Network attack resulting in the appearance of a new equipment on the industrial network or malformed packet (called Scenario 1 in the following subsection),
- Process attack targeting physical components. They affect one or more PLCs and degrade the Operative part (Scenarios 2, 3 and 4 in the following subsections).

These scenarios can be situated with reference to the MITRE ATT&CK for ICS framework [11] as specified in Table 1.

In both cases (network or process attacks), the attacks are generated from Python scripts contained in the Attack Generator VM and are then reverberated onto the testbed. The scripts allow to choose the attack scenario (scenarios from 1 to 4 are described in the following section) and select some parameters depending on the chosen attack. Each scenario can be deployed into different variations.

In the case of a Network attack, there is a single variation available concerning the network to target. The

2. <https://zeek.org/>

TABLE 1. MAPPING OF OUR ATTACK SCENARIOS TO THE MITRE ATT&CK FRAMEWORK

MITRE ATT&CK Framework Attack Scenario	Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Control	Process	Impact
Network Attack	Rogue Master, Supply Chain Compromise	Scripting	none	none	none	Network Sniffing	none	Automated Collection	Commonly Used Port	none	none		Theft of Operational Information
Process Attack	Supply Chain Compromise	Modify Controller Tasking	Modify Program	none	none	none	Program Download	none	Commonly Used Port	Manipulate I/O Image	Brute Force I/O, Modify Parameter, Spoof Reporting Message, Unauthorized Command Message		Manipulation of Control, Manipulation of View, Damage to Property

network attack can either target the industrial or the fieldbus network (further detailed).

In the case of process attacks, several variations exist and 3 parameters can be changed: the targeted Domain, the choice of fooling Supervisory Control or not, and the attack vector. First of all, process attack scenarios can operate either on Propulsion or Direction domain. Concerning the second point, fooling Supervisory Control is an option that allows to hide the effect of the attack and send false data to the Supervisory Control. The effect of this parameter is illustrated in Fig. 6, where the impacts on process components are illustrated depending on the case. The third parameter permits to choose between two distinct attack vectors. The first attack vector is a corrupted PLC containing dead code in its control law (possibly added during a maintenance phase, via a supply chain attack). In this case, the Attack Generator VM scripts allows to select the PLC to target. Then, a bit is written in memory of the targeted PLC which will trigger the dead code. The second attack vector is the fieldbus. In that case, the Attack Generator VM executes scripts making command injection directly on the fieldbus through one of the RS-485 to USB converter.

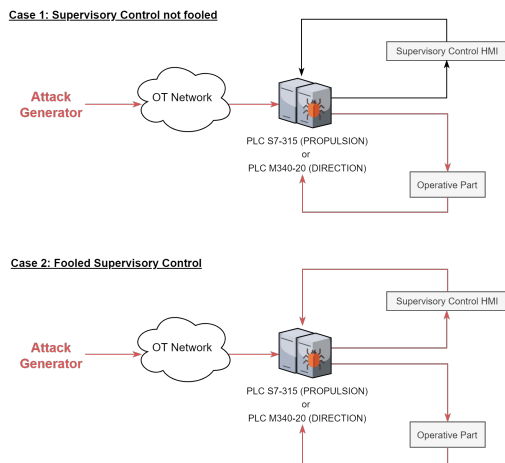


Figure 6. Process Attacks Executed on our Testbed (in red: the impact of the attacks on process components)

4.2.1. Scenario 1: Connection of an Unknown Equipment. This scenario consists in generating the appearance of a new equipment (new node) on the industrial network of the testbed. Traffic is generated on the network as the

new node pings several assets of the network. On the TCP network, this scenario is detected by the OT IDS network probes. On the fieldbus network, since one RS-485 probe is used to inject malicious command, the other is used for detection.

4.2.2. Scenario 2: Violation of a Security Invariant.

This scenario consists in violating a security invariant on the testbed. A safety invariant is an event that must be verified regardless of the operational context of the vessel. For example, in the Direction Domain the position of the right and left rudders must be equal. This scenario forces a mirroring of the rudders, that would cause the steering system to break in a real case. For the purpose of the demonstration, the rudder angles used are deliberately exaggerated as shown in Fig. 7. On the TCP network, this scenario is detected with OT IDS sensors via tailored settings and also with behavioral models. On the fieldbus network, the scenario is also detected by our detection modules.



Figure 7. Impact of Scenario 2 on the Rudders

4.2.3. Scenario 3: Offset Attack. This scenario consists in generating a phenomenon of amplitude or temporal disturbance during a change of setpoint (specific modification of the desired amplitude or duration of application of an order) on the testbed. When it is triggered, it has the effect of modifying the setpoint value in a specific way, violating the previously established sequence of control events, or modifying the duration of application of the setpoint value, violating the previously established time sequence of control events. As for Scenario 2, depending on which network the attack was launched, the attack can be detected by the OT IDS sensors or the fieldbus probes. In the case of an attack without fooling the Supervisory Control (case 1 of Fig. 6), the detection is done with behavioral models based on the information collected by IDS OT sensors when its Deep Packet Inspection (DPI) process allows access to the process values. In the case of an

attack fooling Supervisory Control (case 2 of Fig. 6), the detection can only be done via fieldbus probes. Therefore, the detection is not ensured for the Propulsion domain (since no fieldbus probes are implemented in this domain yet).

4.2.4. Scenario 4: Cavitation Attack. This scenario consists in generating a cavitation phenomenon (oscillation around an operating point) on the testbed. As for the other process attacks, depending on the network level chosen for the attack and on the choice of fooling Supervisory Control or not, the detection is possible through IDS OT sensors or fieldbus sensors.

4.3. Summary

Through the attack scenarios presented in the previous section, the testbed allows to generate a large diversity of threats (network and process) and attack paths (with bounces, a prior supply chain attack, etc.) as well as at different levels of the Purdue model (from Level 5 to 1). Thus, it allows to study how the attack propagates from one level to another (lateral movement). Moreover, it allows to instantiate different solutions (physical and virtual) at different levels and of different natures (probe, firewall, behavioral models, etc.). Finally, it allows to generate representative datasets in order to quickly prototype solutions that can be easily integrated thanks to the flexibility of our testbed's architecture.

5. Comparison with other relevant testbeds

Once again, since [8] is a recent and complete review, we will focus only in this section on testbeds and CRs which are relevant to our work, i.e. those which are dedicated to maritime domain, or those not cited by [8] but which have technical proximities with our work.

Reference [12] proposes a roadmap for the creation of a specialized maritime-cyber lab, which combines maritime technology and traditional cyber-security labs. The proposed Cyber-SHIP Lab (Hardware, Software, Information and Protections) hosts a range of real, non-simulated, maritime systems. This paper mainly describes the planned future works on Cyber-SHIP platform, e.g. gathering real maritime equipment that can be found on a ship's bridge for pentesting, providing signal simulation to test spoofing and jamming and software tools for vulnerability and risk analysis. Even if it is impossible to recreate every available bridge setups, [12] claims that a number of bridge configurations can be achieved with a single lab. The authors admit that the main drawback of Cyber-SHIP Lab is its prohibitive cost. Moreover, many technical details are lacking in this paper, e.g., the detailed architecture of the Lab, its IT/OT components, the covered domain loops of the ship and the considered cyber-attacks. It is thus difficult to compare our work with this reference.

The continuation of this work is described in [13], where the desired platform depicted in [12] is still not completed at the publishing time. Reference [13] provides, a conceptual and generic analysis on how CRs can be used in the maritime context. It presents a hybrid approach combining real physical data and simulation, where up to 3 physical testbeds can be interconnected. The architecture

of their CR is simplified into 3 layers: OT part tends to exist on the physical layers (e.g., propulsion, energy, etc.), while IT part lies on the higher layers (e.g., network packets, operational data). Even interesting, only a few technical details on the CR are provided, for example it is not easy to differentiate what is really physical from virtual, and the available attacks and detection means are not described.

Reference [14] proposes a virtual port logistics and supply chain cybersecurity training platform. This CR is built for organizing small cyber exercises on specific themes. It is hosted on a rack server, the network side is virtualized, and the computers used in the exercises are running in VMs. It enables the simulation of cyberattacks and defences, identification of threats and vulnerabilities, traffic monitoring and in-depth analysis. It consists mainly of customized workflows for setting up the scenarios with virtualized systems. Participants use a remote desktop or similar for connecting to the environment. Technical vs. non-technical information can be displayed during the trainings, depending on the technical level competencies of the trainees. It is difficult to fairly compare our approach with this one since technical details are lacking in [14]. However, we can guess that this testbed is mainly virtual, contrary to ours. OT components can be optionally included in their CR, while our testbed is mainly dedicated to this kind of devices. Moreover, their trainees have only visibility (and then, can only act) on user scenarios and scenario VMs, and not on network topology, virtual and hardware environment. Finally, the available attacks are not described in details, it is only mentioned that they are mainly scripts that execute malicious files or inject traffic into the system.

Reference [15] presents a maritime CR environment, combining navigational, information and telecommunications systems, networks, and SCADA systems. It supports maritime vulnerability, penetration and exploitation scenarios, traffic eavesdropping, positional systems spoofing, navigation takeover, signal intelligence scenarios, and others. It combines simulated and emulated systems to be as realistic as possible. It is made of two main entities: a single ship or a fleet, and an ashore maritime center. Among all the implemented maritime CR components, the Machinery Control System (MCS) is the most related to our work, since it involves PLCs to get information about the ship's machines characteristics (e.g., temperature, engine speed, fuel supply, etc.). The work presented in [15] is particularly focused on radar and AIS attacks, contrary to ours on ICSs. Moreover, no detail is provided on the ICS layer, making comparisons difficult. Finally, it is claimed that various hacking positions are possible, but no precise attack scenario is given. It is difficult to both compare with [15] and assess the feasibility of their attacks.

Reference [16] presents a testbed for cyber security awareness in the maritime domain. Two subsystems are implemented: the propulsion system with the engine control, and the navigation system with a rudder. Some sensors and actuators (e.g., temperature sensing) are simulated with Arduinos. The testbed instantiates also the Level 2 of the Purdue model (see Fig. 1) with a SCADA component implemented with a dedicated HMI. IDSs are also deployed to check some basic safety properties. Three

attacks are described: forging a TCP packet with a specific payload to stop the program loop of one automaton, using specific messages to write data to the memory of one automaton, combining two situations that are considered valid individually, but where their conjunction leads to an unfortunate situation. It appears that it is a small scale ICS testbed, with a small number of integrated components, and only two implemented domains of a warship. Moreover, it is not completely physical, so its fidelity can be considered lower than our testbed.

In our opinion, even not addressing the maritime domain in particular, the nearest work from ours is described in [17]. In this paper, two demonstrators are shown: an Advanced Persistent Threat (APT) demonstrator used for awareness training (called WonderICS) and a flexible lab used for students training and pentesting (called G-ICS). Both testbeds are based on a common concept called Hardware-In-the-Loop (HIL) which combines the advantages of virtualization and real CRs. The goals of the authors are to: i) focusing on getting realistic behaviors from the Level 1 of the Purdue model (see Fig. 1), by integrating real field devices like PLCs; ii) virtualizing the other components, like the attacks generation process through VMs, and the physical process. False data injection and man-in-the-middle-attacks can be launched on demand. WonderICS testbed's scale is equivalent to ours, while G-ICS is bigger (more than 100 industrial devices). Contrary to our work, the physical process is virtualized alongside sensors and actuators.

6. Conclusion and Perspectives

In this paper, we presented an Industrial Control System physical testbed dedicated to naval defense cybersecurity research. This testbed, instantiating real industrial components and networks at the lowest levels of the Purdue model, allows safe, precise and intensive study of the effects of propagating attacks and corresponding detection means. Four different attack scenarios and related countermeasures linked to the MITRE ICS ATT&CK framework have illustrated the technical relevance of our testbed. This latter is also used for indoor exhibitions and trainings. Before our work, only a few notable effort had been made to implement testbeds in the maritime context. In our opinion, our ICS physical testbed appears to be one of the most advanced of the state-of-the-art. Additionally, it is combined with a MIS (VMs) allowing to have a complete industrial system, with a high level of flexibility and repeatability.

Our future works will go in three directions. First, enhancing physical testbed capabilities is a continuous work in progress: we will use the fact that our testbed is easily expandable by integrating, testing and validating new industrial components, and communication and positioning systems, especially to assess the effect of spoofing and jamming radio attacks. Second, we will implement new attack scenarios for a broader MITRE ICS ATT&CK coverage, and develop an even more user-friendly interface to automatically launch attacks. Finally, this testbed will help us to benchmark innovative cyberdefense measures (like knowledge-based and behavior-based intrusion detection) through the generation of representative datasets.

Acknowledgment

The authors would like to thank the anonymous reviewers for interesting feedback, and our colleagues Nidhal Ben Aloui (Naval Group, Naval Cyber Laboratory) for initiating the testbed, and Jean-Baptiste Paux (Naval Group, Naval Cyber Laboratory) for his continuous help in improving our testbed.

References

- [1] K. Stouffer, V. Pillitteri, S. Lightman, and M. Abrams, "Guide to Industrial Control Systems (ICS) Security," NIST Special Publication 800-82 Revision 2, Tech. Rep., 2015.
- [2] N. Falliere, L. O. Murchu, and E. Chien, "W32. stuxnet dossier," *White paper, symantec corp., security response*, vol. 5, no. 6, p. 29, 2011, <https://pax0r.com/hh/stuxnet/Symantec-Stuxnet-Update-Feb-2011.pdf>.
- [3] R. Khan, P. Maynard, K. McLaughlin, D. Lavery, and S. Sezer, "Threat Analysis of BlackEnergy Malware for Synchrophasor Based Real-time Control and Monitoring in Smart Grid," in *BCS International Symposium for ICS & SCADA Cyber Security Research (ICS-CSR)*, 2016, pp. 53–63.
- [4] "Recommendations Following the Oldsmar Water Treatment Facility Cyber Attack," Dragos Inc., Technical Report, 2021.
- [5] A. Drougkas, A. Sarri, and P. Kyranoudi, "Cyber Risk Management for Ports – Guidelines for Cybersecurity in the Maritime Sector," ENISA, Tech. Rep., 2020.
- [6] N. Pelissero, P. M. Laso, O. Jacq, and J. Puentes, "Towards Modeling of Naval Systems Interdependencies for Cybersecurity," in *IEEE OCEANS*, 2021, pp. 1–7.
- [7] "IEC 62264: Enterprise-control system integration," Industrial Electrotechnical Commission, Technical Specification, 2010.
- [8] M. Conti, D. Donadel, and F. Turrin, "A Survey on Industrial Control System Testbeds and Datasets for Security Research," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 4, pp. 2248–2294, 2021.
- [9] D. Holmes, M. Papathanasakis, L. Maglaras, M. A. Ferrag, S. Nepal, and H. Janicke, "Digital Twins and Cyber Security – Solution or Challenge?" in *IEEE South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)*, 2021, pp. 1–8.
- [10] "IEC 62264: Enterprise-Control System Integration," Industrial Electrotechnical Commission, Technical Specification, 2010.
- [11] T. M. Corporation. (2021) ATT&CK® for Industrial Control Systems. [Online]. Available: https://collaborate.mitre.org/attackics/index.php/Main_Page
- [12] K. Tam and K. Jones, "Cyber-SHIP: Developing Next Generation Maritime Cyber Research Capabilities," in *University of Plymouth Pedagogic Research and Development Database*, 2019.
- [13] K. Tam, M.-N. Kemedi, and K. Jones, "The Use of Cyber Ranges in the Maritime Context: Assessing Maritime-Cyber Risks, Raising Awareness, and Providing Training," *Maritime Technology and Research*, vol. 3, no. 1, pp. 16–30, 2021.
- [14] H. Pyykkö, J. Kuusijärvi, S. Noponen, S. Toivonen, and V. Hinkka, "Building a Virtual Maritime Logistics Cybersecurity Training Platform," in *Hamburg International Conference of Logistics (HICL)*, vol. 29, 2020, pp. 223–246.
- [15] G. Potamos, A. Peratikou, and S. Stavrou, "Towards a Maritime Cyber Range Training Environment," in *IEEE International Conference on Cyber Security and Resilience*, 2021, pp. 180–185.
- [16] T. Becmeur, X. Boudvin, D. Brosset, G. Héno, T. Merien, O. Jacq, Y. Kermarrec, and B. Sultan, "A Platform for Raising Awareness on Cyber Security in a Maritime Context," in *IEEE International Conference on Computational Science and Computational Intelligence (CSCI)*, 2017, pp. 103–108.
- [17] M. Puys, P.-H. Thevenon, and S. Mocanu, "Hardware-In-The-Loop Labs for SCADA Cybersecurity Awareness and Training," in *ACM International Conference on Availability, Reliability and Security (ARES)*, 2021, pp. 1–10.